Problem 1

Game 1

$$\int_1^4 \{x\}^{\lfloor x \rfloor} \, \mathrm{d}x$$

The UK University Integration Bee

2024/25

Chainlink Round

Round 3

Problem 2

Game 1

$$\int \frac{\sinh(x) + \cosh(x)}{\sinh(e^x) + \cosh(e^x)} \, \mathrm{d}x$$

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Round 3

Problem 3

Game 1

$$\int \frac{1 + \ln(x)}{\sqrt{x^{-2x} - 1}} \, \mathrm{d}x$$

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Round 3

Problem 4

Game 1

$$\int_1^\infty \frac{1}{x\sqrt{x}\sqrt{x-1}} \, \mathrm{d}x$$

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Round 3

Problem 5

Game 1

$$\int_0^1 \frac{\ln(1+x+x^2)}{x}\,\mathrm{d}x$$

Problem 6

Game 1

$$\int_0^1 f(x)\,\mathrm{d}x$$

$$\text{where } f(x) = \begin{cases} 1, & x = 0.x_1x_2x_3\ldots \text{ where } x_n \neq 9 \text{ for all } n; \\ 0, & \text{else} \end{cases}$$

Problem 7

Game 1

$$\int_{\tan^{-1}\left(\frac{1}{3}\right)}^{\tan^{-1}\left(\frac{1}{2}\right)} \frac{x}{\sin(4x)}\,\mathrm{d}x$$

Problem 8

Game 1

$$\lim_{n\to\infty}(n+1)\int_0^1 x^{x^x+n}\ln(1+x+x^x+x^{x^x}+x^{x^{x^x}})\,\mathrm{d}x$$